Question

V_2O_5 and S_2Cl_2 react at a 1:3 ratio at high temperature to produce three substances.

Select the correct option regarding the balanced chemical reaction equation.

A. All other options are incorrect

B. The product of all coefficients in a chemical reaction equation is 108.

C. The product of all coefficients in a chemical reaction equation is 1860.

D. The molar ratio of sulfur-containing products in the reaction products is 1:1

E. Vanadium-containing reaction product partial hydrolysis will generate vanadium-oxygen double bonds.

F. The coefficient of vanadium pentoxide in the reaction equation is 1.

Answer

Correct Answer: A

Detailed Explanation

Vanadium pentoxide has strong oxidizing properties and can oxidize +1 valent sulfur to sulfur dioxide at high temperatures. With a reactant ratio of 1:3, vanadium is insufficient to oxidize all sulfur to sulfur dioxide, inevitably leaving some sulfur in a lower valence state, the most reasonable form of which is elemental sulfur. Disulfur dichloride can act as a chlorinating agent, so the products should be vanadium chloride, sulfur dioxide, and elemental sulfur.

Among vanadium chlorides, VCl_3 and VCl_4 are relatively stable. Since there is an excess of low-valent sulfur present here, the lower valence state VCl_3 is more reasonable.

CHECKPOINT 1 PTS

The products are SO_2, VCl_3, and S.

The balanced chemical reaction equation is:

$$2V_2O_5 + 6S_2Cl_2 \rightarrow 4VCl_3 + 5SO_2 + 7S$$

CHECKPOINT 2 PTS

$$2V_2O_5 + 6S_2Cl_2 \rightarrow 4VCl_3 + 5SO_2 + 7S$$

The product of all the coefficients in the chemical reaction equation is 1680. Therefore, options B and C are incorrect.

The molar ratio of sulfur-containing products is 5:7. Therefore, option D is incorrect.

Partial hydrolysis of vanadium trichloride cannot generate vanadium-oxygen double bonds. Therefore, option E is incorrect.

CHECKPOINT 1 PTS

Partial hydrolysis of VCl_3 cannot generate vanadium-oxygen double bonds

The coefficient of vanadium pentoxide is 2. Therefore, option F is incorrect.

There is no correct option, so choose option A.